FINANCIAL RISK & LOAN DEFAULT ANALYTICS

Project Report

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Tools: Python • SQL • Power BI • Excel/CSV

32,416 Total loan applications	**7,089*** Default loans
21.87% Displayed default rate	**■9,593.85** Average loan amount

***Default-loan count and default-rate figures should be reconciled against the final cleaned dataset and Power BI measures before external publication.**

A portfolio project demonstrating a practical analytics workflow: data understanding, data cleaning, SQL analysis, dashboarding, and communication of risk patterns.

1. Executive Summary

This project examines loan application data to explore patterns associated with loan defaults and to present portfolio-level indicators in an interactive Power BI dashboard. The workflow combines Python-based data preparation and exploratory analysis, SQL-based summaries, and dashboard reporting.

The dashboard is branded as **Skyline Bank** for presentation purposes. Skyline Bank is a fictional brand used only to demonstrate dashboard design; this report does not represent a real financial institution.

The main objective is to help a reader explore how observed default rates vary across loan grades and applicant characteristics. The analysis is descriptive and does not establish causation or predict the behavior of future borrowers.

2. Business Problem & Objectives

Lenders need clear ways to monitor loan portfolios and understand where defaults are more common. This project translates loan-level records into summary metrics and category-level comparisons that can support exploratory review.

Objectives

• Inspect the dataset and assess data quality.
• Prepare a cleaned dataset for analysis.
• Summarize loan applications and default outcomes using SQL.
• Build an interactive Power BI dashboard with KPI cards, charts, and slicers.
• Communicate observed differences across loan grades and home-ownership categories.

3. Dataset Overview

The dataset contains 32,416 records after cleaning and 12 columns. Fields include applicant age, income, employment length, home ownership, loan purpose, loan grade, loan amount, interest rate, loan status, loan-to-income ratio, prior-default indicator, and credit-history length.

Field group	Examples	Analytical use
Applicant profile	Age, income, employment length, home ownership	Compare applicant segments
Loan characteristics	Amount, interest rate, purpose, grade	Explore portfolio composition and grade-level patterns
Credit history	Prior-default indicator, credit-history length	Review historical credit-related segments
Outcome	Loan status (0/1)	Calculate observed default counts and rates

Data-quality note: The recorded cleaning workflow identified missing values in employment length and interest rate, along with duplicate records. The cleaned output was reported as having no missing values and no duplicates. The exact imputation and duplicate-removal rules should be documented in the notebook.

4. Tools & Methodology

Python: Used for data inspection, cleaning, feature preparation, and exploratory analysis. The reported cleaned dataset contains 32,416 rows and 12 columns.

SQL: Used to query the loan_applications table and produce summary counts and grouped risk metrics.

Power BI: Used to create an interactive dashboard with KPI cards, visual comparisons, and slicers. The dashboard includes a Skyline Bank visual identity for portfolio presentation.

Excel/CSV: Used for data storage and inspection during the project workflow.

Suggested reproducibility checks: Confirm the final CSV path, verify row counts after cleaning, and ensure SQL and Power BI use the same cleaned dataset.

5. Dashboard KPIs

KPI	Displayed value	Interpretation / validation
Total loan applications	32,416	Confirm with row count in the final cleaned table.
Default loans	7,089*	Reconcile with COUNT where loan_status = 1.
Default rate	21.87%*	Should equal default loans divided by total applications, formatted as percentage.
Average loan amount	■9,593.85	Confirm using the average of loan_amnt in the cleaned data.

***The default count and rate shown above reflect the dashboard figures supplied for this report draft. Earlier data inspection reported a different default count, so validate both figures before publishing the final report.**

6. Exploratory Findings

The initial category summaries showed differences in observed default rates across loan grades and home-ownership categories. These results are descriptive of the dataset and should be interpreted alongside category sample sizes.

Loan grade examples from the exploratory output

Loan grade	Loans	Defaults	Observed default rate
G	64	63	98.44%
F	241	170	70.54%
E	963	621	64.49%
D	3,620	2,138	59.06%
C	6,438	1,336	20.75%
B	10,387	1,695	16.32%

The G-grade percentage is based on only 64 records, so it should not be interpreted as a stable estimate for all borrowers. Grade A was not included in the visible summary available for this report draft.

Home-ownership examples from the exploratory output

Home ownership	Loans	Defaults	Observed default rate
RENT	16,378	5,177	31.61%
OTHER	106	33	31.13%
MORTGAGE	13,369	1,687	12.62%
OWN	2,563	192	7.49%

These category comparisons do not prove that a particular characteristic causes default. Other factors and interactions may influence the observed outcomes.

7. Limitations & Next Steps

Limitations

• The report uses descriptive analysis of the available dataset.

• Data quality and cleaning choices can affect results.
• Category-level rates can be unstable when group sizes are small.
• No validated predictive model or causal analysis is claimed in this report.

Recommended next steps

1. Reconcile default counts and rates across Python, SQL, and Power BI.
2. Document missing-value treatment and duplicate-removal rules.
3. Add a dashboard screenshot and final report PDF to the reports folder.
4. Publish the notebook, SQL scripts, and Power BI file in the project repository, subject to dataset licensing and file-size limits.
5. If extending to machine learning, use a separate train/test workflow, check class imbalance, evaluate with suitable metrics, and avoid presenting an unvalidated model as a lending decision tool.

8. Conclusion

This project demonstrates an end-to-end data analytics workflow for exploring loan portfolio and default patterns. Combining Python, SQL, and Power BI helps turn raw loan records into structured summaries and an interactive dashboard. The project is suitable as a portfolio demonstration once the KPI figures and analysis outputs have been reconciled and documented.

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Disclaimer: This is an educational portfolio project. Skyline Bank is a fictional brand used for presentation only. Findings are descriptive and are not financial advice or a basis for real lending decisions.